Data Mining Project Business Report

DSBA

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Part 1: Clustering: Actionable Insights & Recommendations
- Extract meaningful insights (atleast 3) from the clusters to identify the most effective types of ads, target audiences, or marketing strategies that can be inferred from each segment. - Based on the clustering analysis and key insights, provide actionable recommendations (atleast 3) to Ads24x7 on how to optimize their digital marketing efforts, allocate budgets efficiently, and tailor ad content to specific audience segments.

Part 2: PCA: Define the problem and perform Exploratory Data Analysis
- Problem Definition - Check shape, Data types, statistical summary - Perform an EDA on the data to extract useful insights Note: 1. Pick 5 variables out of the given 24 variables below for EDA: No_HH, TOT_M, TOT_F, M_06, F_06, M_SC, F_SC, M_ST, F_ST, M_LIT, F_LIT, M_ILL, F_ILL, TOT_WORK_M, TOT_WORK_F, MAINWORK_M, MAINWORK_F, MAIN_CL_M, MAIN_CL_F, MAIN_AL_M, MAIN_AL_F, MAIN_HH_M, MAIN_HH_F, MAIN_OT_M, MAIN_OT_F 2. Example questions to answer from EDA - (i) Which state has highest gender ratio and which has the lowest? (ii) Which district has the highest & lowest gender ratio?

Part 2: PCA: Data Preprocessing

- Check for and treat (if needed) missing values - Check for and treat (if needed) data irregularities - Scale the Data using the z-score method - Visualize the data before and after scaling and comment on the impact on outliers.

Part 2: PCA: PCA

- Create the covariance matrix - Get eigen values and eigen vectors - Identify the optimum number of PCs - Show Scree plot - Compare PCs with Actual Columns and identify which is explaining most variance - Write inferences about all the PCs in terms of actual variables - Write linear equation for first PC Note: For the scope of this project, take at least 90% explained variance.

Part 1: Clustering: Define the problem and perform Exploratory Data Analysis

Problem definition- Check shape, Data types, statistical summary- Univariate analysis- Bivariate analysis- Key meaningful observations on individual variables and the relationship between variables.

Based on the data of a Digital Marketing company by their Marketing Intelligence team, it is required to segment type of ads based on the features provided. Use Clustering procedure to segment ads into homogeneous groups.

The following three features are used in digital marketing:

CPM = (Total Campaign Spend / Number of Impressions) * 1,000
CPC = Total Cost (spend) / Number of Clicks
CTR = Total Measured Clicks / Total Measured Ad Impressions x 100

BASIC INFORMATION ABOUT DATASET:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 23066 entries, 0 to 23065
Data columns (total 19 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   Timestamp              23066 non-null  object
 1   InventoryType          23066 non-null  object
 2   Ad - Length            23066 non-null  int64
 3   Ad- Width              23066 non-null  int64
 4   Ad Size                23066 non-null  int64
 5   Ad Type                23066 non-null  object
 6   Platform               23066 non-null  object
 7   Device Type            23066 non-null  object
 8   Format                 23066 non-null  object
 9   Available_Impressions  23066 non-null  int64
 10  Matched_Queries        23066 non-null  int64
 11  Impressions            23066 non-null  int64
 12  Clicks                 23066 non-null  int64
 13  Spend                  23066 non-null  float64
 14  Fee                    23066 non-null  float64
 15  Revenue                23066 non-null  float64
 16  CTR                    18330 non-null  float64
 17  CPM                    18330 non-null  float64
 18  CPC                    18330 non-null  float64
dtypes: float64(6), int64(7), object(6)
memory usage: 3.3+ MB
```

In this dataset we have 23066 rows and 19 columns.

And no duplicate enteries are there.

NUMERICAL SUMMARIZATION ABOUT DATASET:

	count	mean	std	min	25%	50%	75%	max
Ad - Length	23066.0	3.851631e+02	2.336514e+02	120.0000	120.000000	300.00000	7.200000e+02	728.00
Ad- Width	23066.0	3.378960e+02	2.030929e+02	70.0000	250.000000	300.00000	6.000000e+02	600.00
Ad Size	23066.0	9.667447e+04	6.153833e+04	33600.0000	72000.000000	72000.00000	8.400000e+04	216000.00
Available_Impressions	23066.0	2.432044e+06	4.742888e+06	1.0000	33672.250000	483771.00000	2.527712e+06	27592861.00
Matched_Queries	23066.0	1.295099e+06	2.512970e+06	1.0000	18282.500000	258087.50000	1.180700e+06	14702025.00
Impressions	23066.0	1.241520e+06	2.429400e+06	1.0000	7990.500000	225290.00000	1.112428e+06	14194774.00
Clicks	23066.0	1.067852e+04	1.735341e+04	1.0000	710.000000	4425.00000	1.279375e+04	143049.00
Spend	23066.0	2.706626e+03	4.067927e+03	0.0000	85.180000	1425.12500	3.121400e+03	26931.87
Fee	23066.0	3.351231e-01	3.196322e-02	0.2100	0.330000	0.35000	3.500000e-01	0.35
Revenue	23066.0	1.924252e+03	3.105238e+03	0.0000	55.365375	926.33500	2.091338e+03	21276.18
CTR	18330.0	7.366054e-02	7.515992e-02	0.0001	0.002600	0.08255	1.300000e-01	1.00
CPM	18330.0	7.672045e+00	6.481391e+00	0.0000	1.710000	7.66000	1.251000e+01	81.56
CPC	18330.0	3.510606e-01	3.433338e-01	0.0000	0.090000	0.16000	5.700000e-01	7.26

-analyzing the timestamp feature and to change the datatype of timestamp from object to datetime.

```
3892      2020-11-5-17
6216       2020-11-3-7
11637     2020-9-14-10
2989       2020-11-6-8
14992    2020-10-24-21
19330    2020-11-14-10
21849    2020-10-19-13
9586       2020-11-4-0
22124     2020-9-21-23
22912      2020-9-3-22
Name: Timestamp, dtype: object
```

*Observations from above analysis and action items.
*From above analysis we have seen that the timestamp feature is in the format : yyyy-mm-dd-hh

- its datatype is object we have to change it to datetime format

AFTER CHANGING ITS DATATYPE TO DATETIME FORMAT:

```
22371   2020-11-09 04:00:00
5076    2020-11-20 04:00:00
22231   2020-10-04 03:00:00
5118    2020-09-05 02:00:00
11889   2020-10-01 20:00:00
22208   2020-11-14 15:00:00
14129   2020-11-02 11:00:00
7545    2020-09-25 04:00:00
13681   2020-10-22 17:00:00
10752   2020-11-05 22:00:00
Name: Timestamp, dtype: datetime64[ns]
```

Now we have seen that the timestamp feature datatype has been changed from object to datetime.

In the below analysis we can see that there are null values present.

```
Timestamp                   0
InventoryType               0
Ad - Length                 0
Ad- Width                   0
Ad Size                     0
Ad Type                     0
Platform                    0
Device Type                 0
Format                      0
Available_Impressions       0
Matched_Queries             0
Impressions                 0
Clicks                      0
Spend                       0
Fee                         0
Revenue                     0
CTR                      4736
CPM                      4736
CPC                      4736
dtype: int64
```

In the above analysis we have seen that there are null values present in CTR,CPM AND CPC feature which we need to treat.

After treating null values:

```
Timestamp                0.000000
InventoryType            0.000000
Ad - Length              0.000000
Ad- Width                0.000000
Ad Size                  0.000000
Ad Type                  0.000000
Platform                 0.000000
Device Type              0.000000
Format                   0.000000
Available_Impressions    0.000000
Matched_Queries          0.000000
Impressions              0.000000
Clicks                   0.000000
Spend                    0.000000
Fee                      0.000000
Revenue                  0.000000
CTR                      0.205324
CPM                      0.205324
CPC                      0.205324
dtype: float64
```

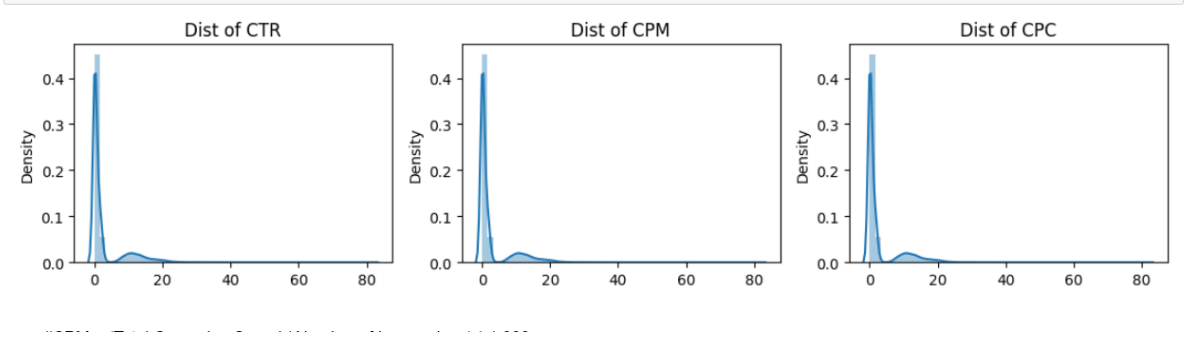
*Missing Values imputation

OBSERVATIONS :

There are 3 Features with nan values. These are- CTR(20% nan/4736), CPM(20% nan/4736) and CTC(20% nan/4736)

We have imputed the missing values with the below given formula:

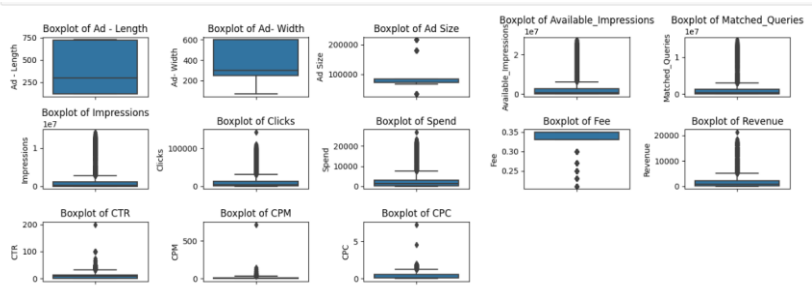
- #CPM = (Total Campaign Spend / Number of Impressions) * 1,000
- #CPC = Total Cost (spend) / Number of Clicks
- #CTR = Total Measured Clicks / Total Measured Ad Impressions x 100



After imputation of null values:

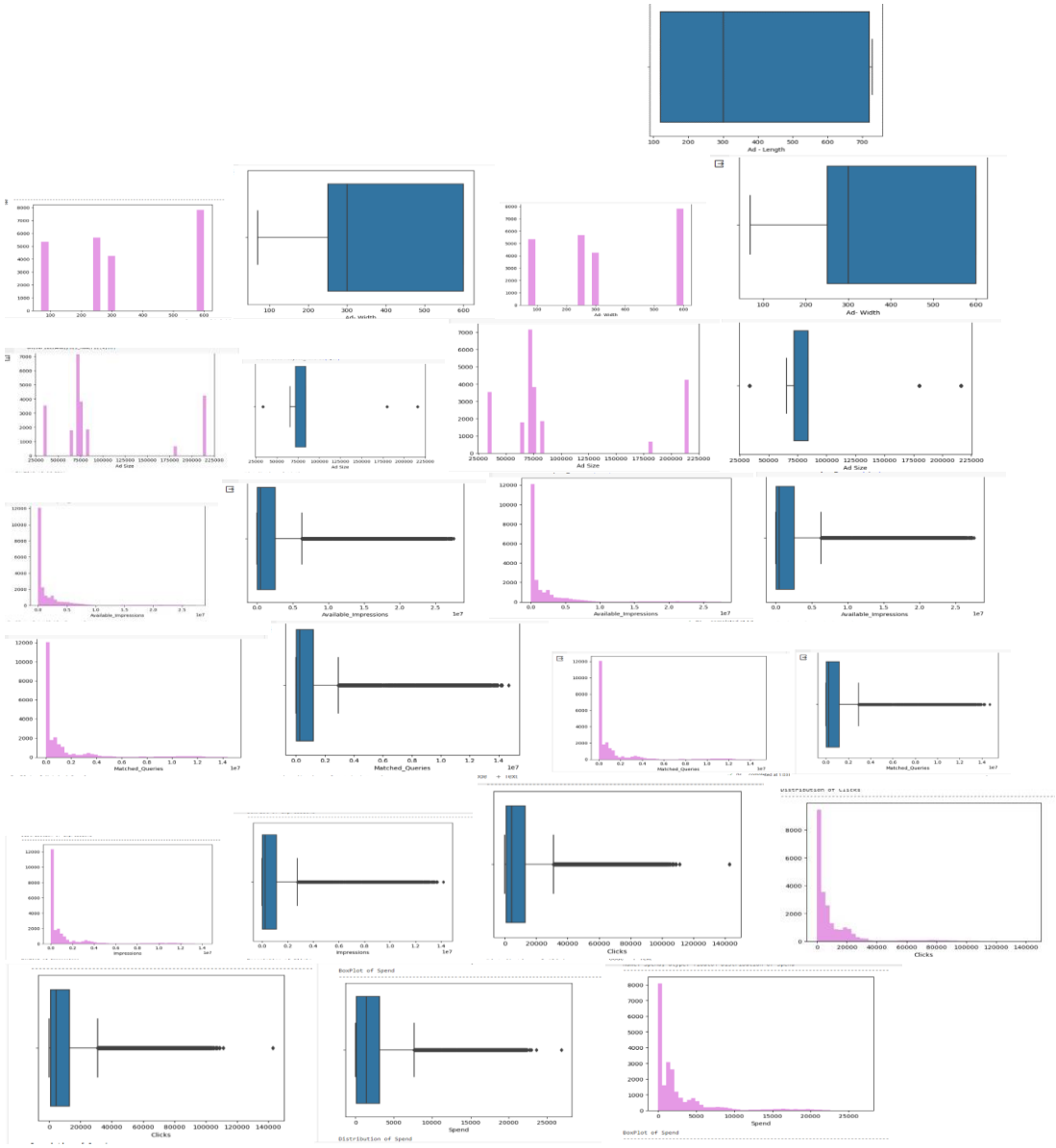
```
Timestamp                0
InventoryType            0
Ad - Length              0
Ad- Width                0
Ad Size                  0
Ad Type                  0
Platform                 0
Device Type              0
Format                   0
Available_Impressions    0
Matched_Queries          0
Impressions              0
Clicks                   0
Spend                    0
Fee                      0
Revenue                  0
CTR                      0
CPM                      0
CPC                      0
dtype: int64
```

Now we have 0 null values present in our dataset after imputing it.



*From the above set of box plots, its evident that Outliers are present in all numeric Features except for Ad-length and Ad-width.

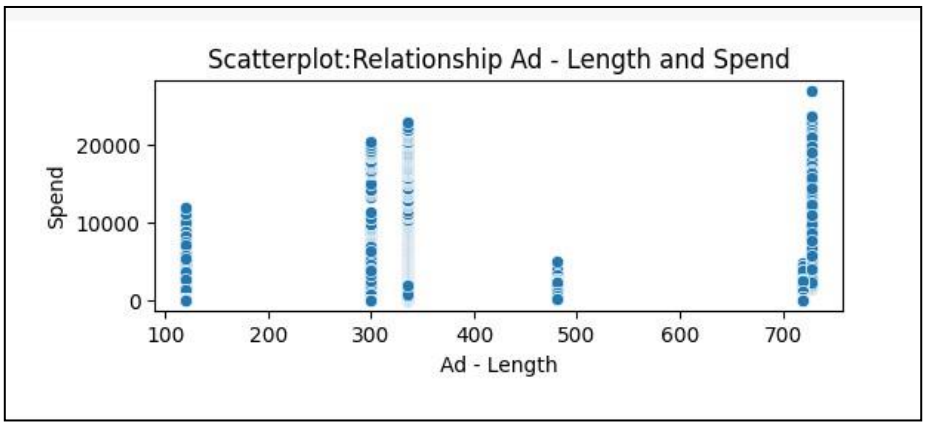
UNIVARIATE ANALYSIS:



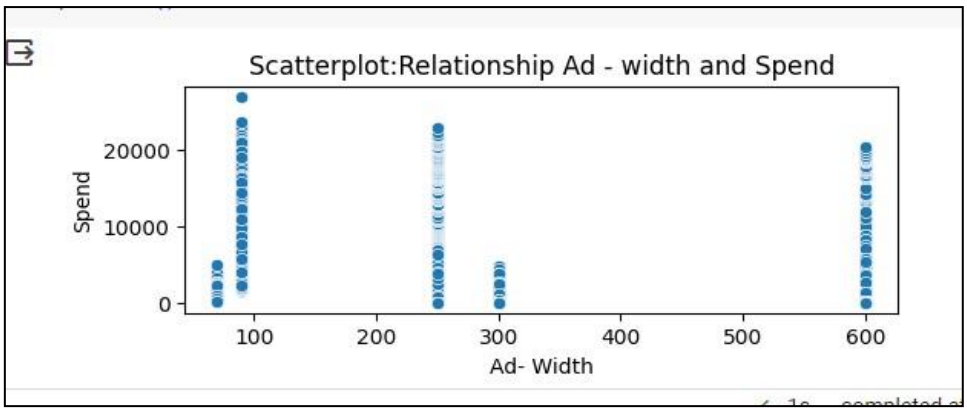
From the univariate analysis we can noticed all boxplots are left skewed.

company using very less amount in their ads and promotions part so they did not earned much revenue . they should be focus on promotion part and also they must be maintain theirad-length. Many people did not like long ads, so ads should be prepared in a short & sweet manner.

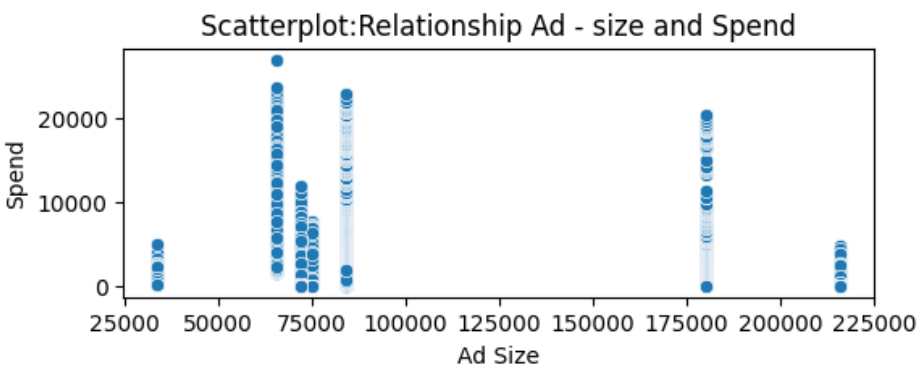
BIVARIATE ANALYSIS:



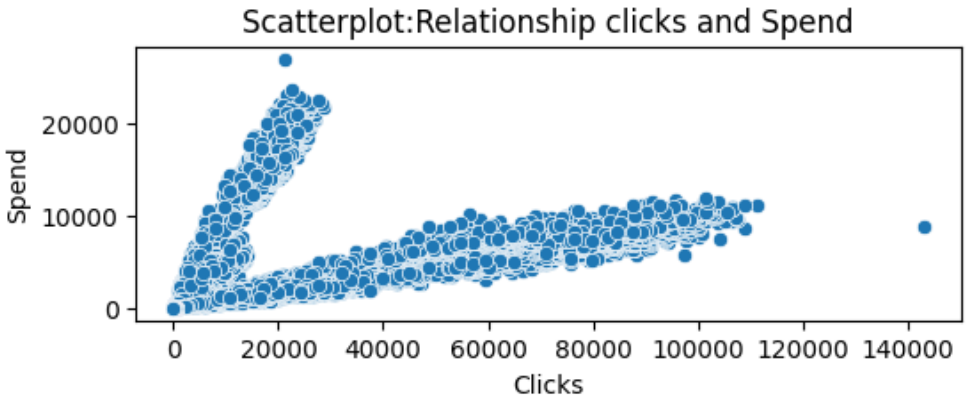
From the above plot we can not see any relationship between spend and ad length.



From the above plot we can not see any relationship between ad width and spend.



from the above graph we can not see any relationship between ad size and spend.



from above scatter plot also we can not find any positive or negative relationship between clicks and spend.

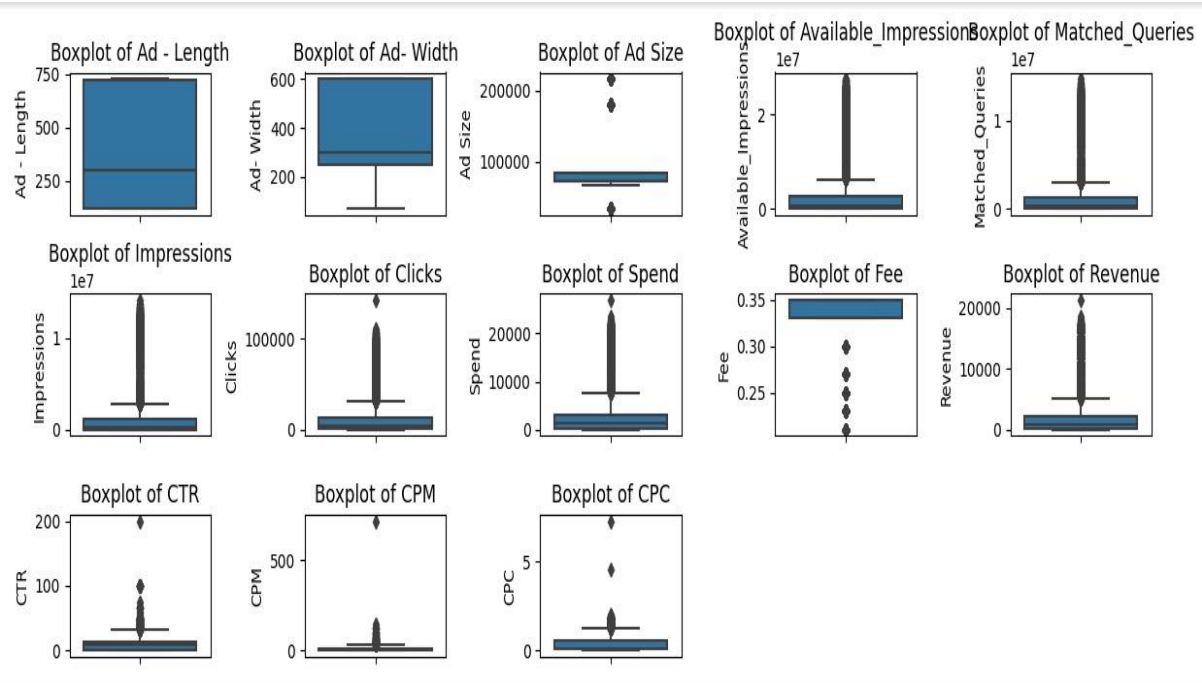
Part 1: Clustering: Data Preprocessing

- Missing value check and treatment- Outlier Treatment- z-score scaling Note: Treat missing values in CPC, CTR and CPM using the formula given.

OUTLIER TREATMENT K-Means algorithm is sensitive to outliers and may not give best results in case outliers are present in the data. The outlier(s) may increases the mean of data significantly and since K-Means algorithm is about finding mean of clusters, the algorithm is influenced by outliers.

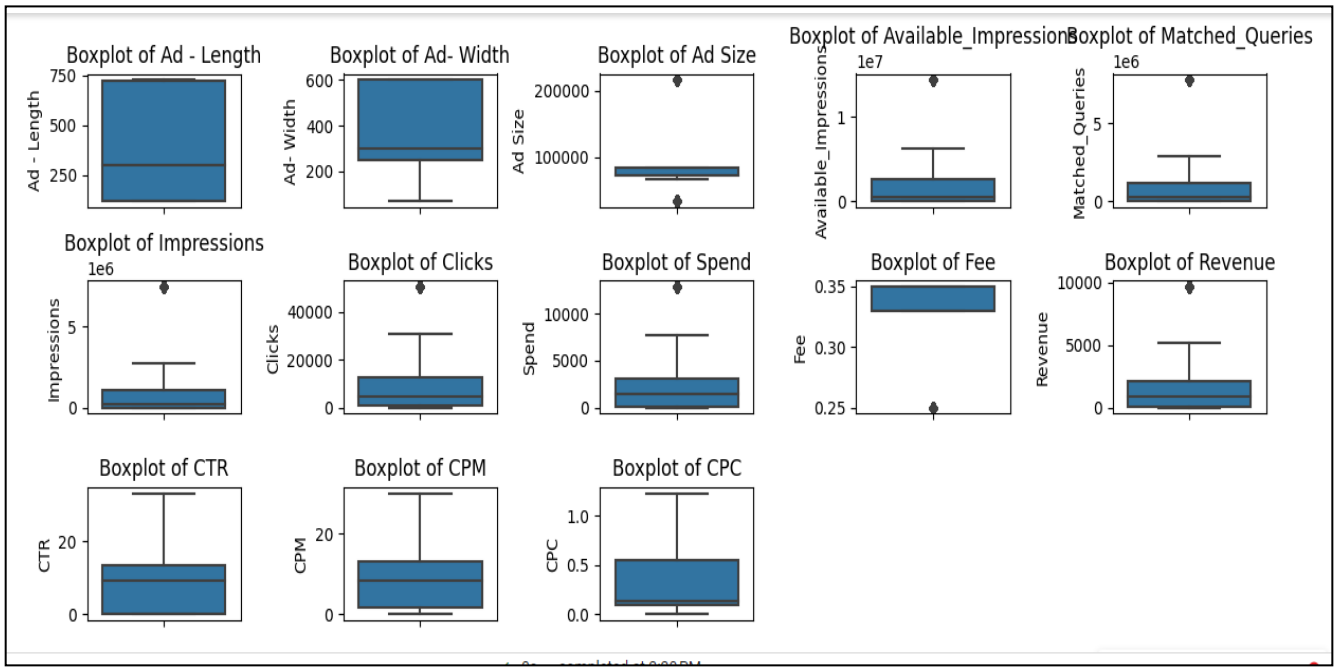
Even if the outliers are present in a low percentage, they can have a significant impact on clustering. Hence it is better to identify and remove outliers before applying K-means clustering algorithm.

BOXPLOT BEFORE TREATING THE OUTLIERS:



We can see in this data many outliers are there, we need treat them before doing theK-means clustering. It is evident that only in Ad- length and Ad-Width no outliers are there.

BOXPLOT AFTER TREATING OUTLIERS WITH IQR:



*SCALING THE DATA:

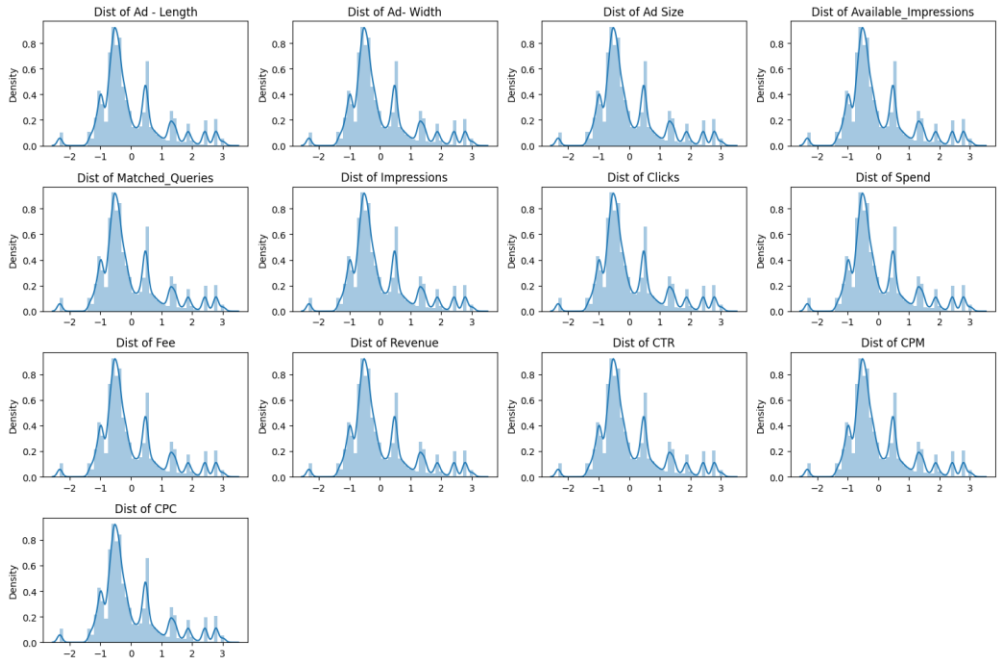
Perform z-score scaling and discuss how it affects the speed of the algorithm. We will perform Z score on the Dataset. Z-score normalization refers to the process of normalizing every value in a dataset such that the mean of all of the values is 0 and the standard deviation is 1.

AFTER THE DATA HAS BEEN SCALED:

Ad - Length	Ad- Width	Ad Size	Available_Impressions	Matched_Queries	Impressions	Clicks	Spend	Fee	Revenue	CTR	CPM	CPC
-0.364496	-0.432797	-0.359227	-0.569484	-0.567061	-0.563943	-0.719779	-0.722776	0.487214	-0.676118	-0.978830	-1.220346	-1.083011
-0.364496	-0.432797	-0.359227	-0.569490	-0.567076	-0.563958	-0.719779	-0.722776	0.487214	-0.676118	-0.973650	-1.220346	-1.083011
-0.364496	-0.432797	-0.359227	-0.569269	-0.567049	-0.563931	-0.719779	-0.722776	0.487214	-0.676118	-0.982332	-1.220346	-1.083011
-0.364496	-0.432797	-0.359227	-0.569339	-0.566994	-0.563875	-0.719779	-0.722776	0.487214	-0.676118	-0.992329	-1.220346	-1.083011
-0.364496	-0.432797	-0.359227	-0.569622	-0.567093	-0.563975	-0.719779	-0.722776	0.487214	-0.676118	-0.965826	-1.220346	-1.083011

Now we can identify that in this dataset the mean of all of the values is 0 and the standard deviation is 1.

DISTRIBUTION OF EACH COLUMN IN THR DATAFRAME:



And in the above plots we can see that all the features follow a normal distribution.

	count	mean	std	min	25%	50%	75%	max
Ad - Length	23066.0	0.0	1.0	-1.13	-1.13	-0.36	1.43	1.47
Ad- Width	23066.0	-0.0	1.0	-1.32	-0.43	-0.19	1.29	1.29
Ad Size	23066.0	-0.0	1.0	-1.01	-0.41	-0.41	-0.22	1.87
Available_Impressions	23066.0	0.0	1.0	-0.57	-0.56	-0.46	0.02	2.78
Matched_Queries	23066.0	-0.0	1.0	-0.57	-0.56	-0.47	-0.11	2.43
Impressions	23066.0	0.0	1.0	-0.56	-0.56	-0.47	-0.12	2.40
Clicks	23066.0	0.0	1.0	-0.72	-0.67	-0.39	0.22	3.02
Spend	23066.0	-0.0	1.0	-0.72	-0.70	-0.33	0.13	2.81
Fee	23066.0	0.0	1.0	-2.32	-0.07	0.49	0.49	0.49
Revenue	23066.0	0.0	1.0	-0.68	-0.66	-0.35	0.07	2.76
CTR	23066.0	-0.0	1.0	-1.02	-0.98	0.16	0.67	3.13
CPM	23066.0	-0.0	1.0	-1.22	-0.96	0.04	0.74	3.28
CPC	23066.0	-0.0	1.0	-1.08	-0.78	-0.61	0.75	3.05

Part 1: Clustering: Hierarchical Clustering

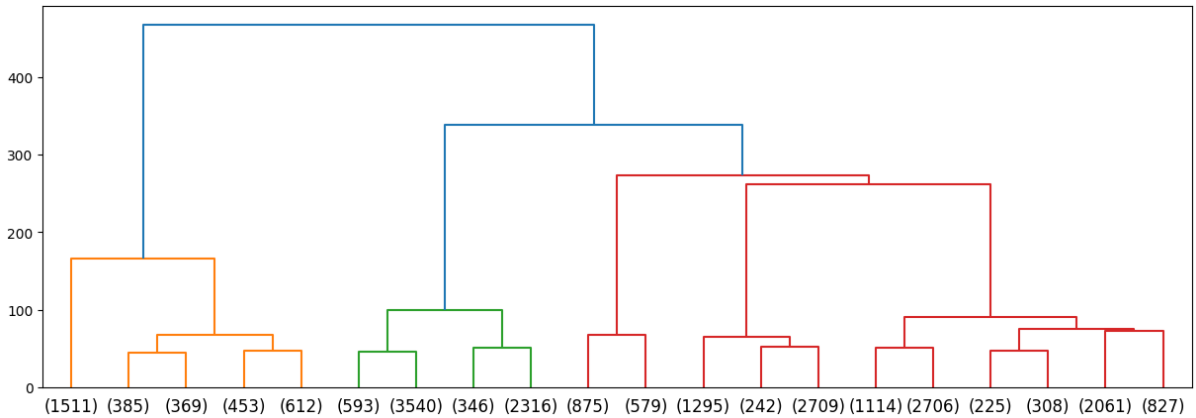
- Construct a dendrogram using Ward linkage and Euclidean distance - Identify the optimum number of Clusters.

Creating Clusters using HIERARCHIAL CLUSTERING

DENDOGRAM USING WARD's LINKAGE

A dendrogram created using the Ward's linkage method is a visual representation of hierarchical clustering results. Ward's linkage is a distance-based algorithm that minimizes the variance within each cluster when merging them.

The dendrogram depicts the hierarchical relationships between data points by using branches and clusters. The height of each branch corresponds to the dissimilarity or distance between clusters. By analyzing the dendrogram, we can determine the optimal number of clusters or levels of granularity in our clustering analysis using Ward's linkage.



NUMBER OF UNIQUE CLUSTERS:

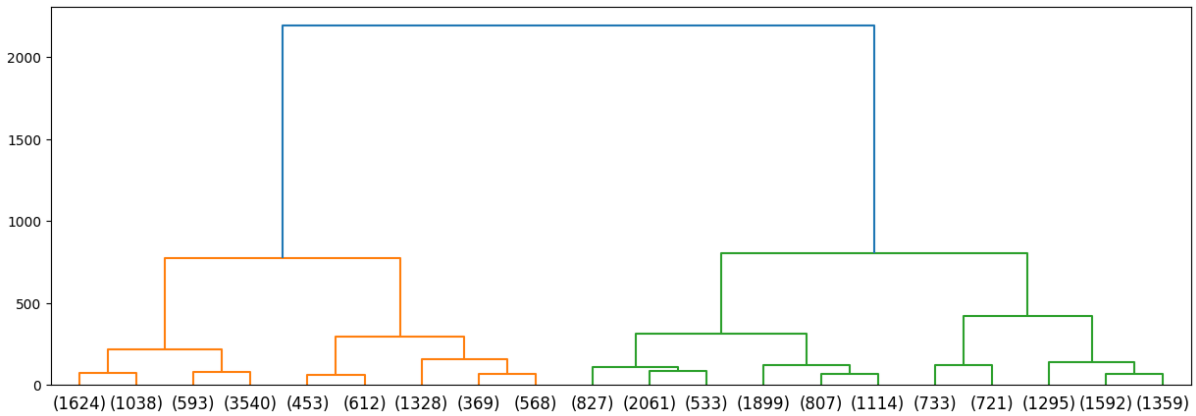
```
array([12, 27, 38, 26, 42, 39, 41, 35, 34, 37, 36, 13, 19, 32, 14, 33, 15,
       31, 29, 40, 18, 30, 16, 28, 25, 22, 17, 11, 10,  9,  7,  8,  5,  6,
       24, 23,  4, 21,  1, 20,  3,  2])
```

DENDOGRAM USING EUCLIDEAN DISTANCES

A dendrogram constructed using the Euclidean distance metric visualizes the hierarchical relationships between data points in a clustering analysis. The Euclidean distance measures the straight-line distance between two data points in a multi-dimensional space.

In the dendrogram, each data point is represented as a leaf, and the branches indicate the similarity or dissimilarity between them based on their Euclidean distances. The height of each branch represents the magnitude of the dissimilarity.

By examining the dendrogram, we can gain insights into the hierarchical structure of the data and make decisions about the appropriate number of clusters or levels of granularity in our clustering analysis using Euclidean distance.



Part 1: Clustering: K-means Clustering

- Apply K-means Clustering- Plot the Elbow curve- Check Silhouette Scores- Figure out the appropriate number of clusters- Cluster Profiling.

After applying K-means:

```
64077.42983865314
```

WSS:

```
[3323749.850212434,
 844094.4525228802,
 497565.4157036493,
 269799.0182079396,
 193975.78599335745,
 147249.50001246395,
 108027.1881854527,
 84242.57939303336,
 73664.46154469169,
 64077.42983865314]
```

ELBOW PLOT

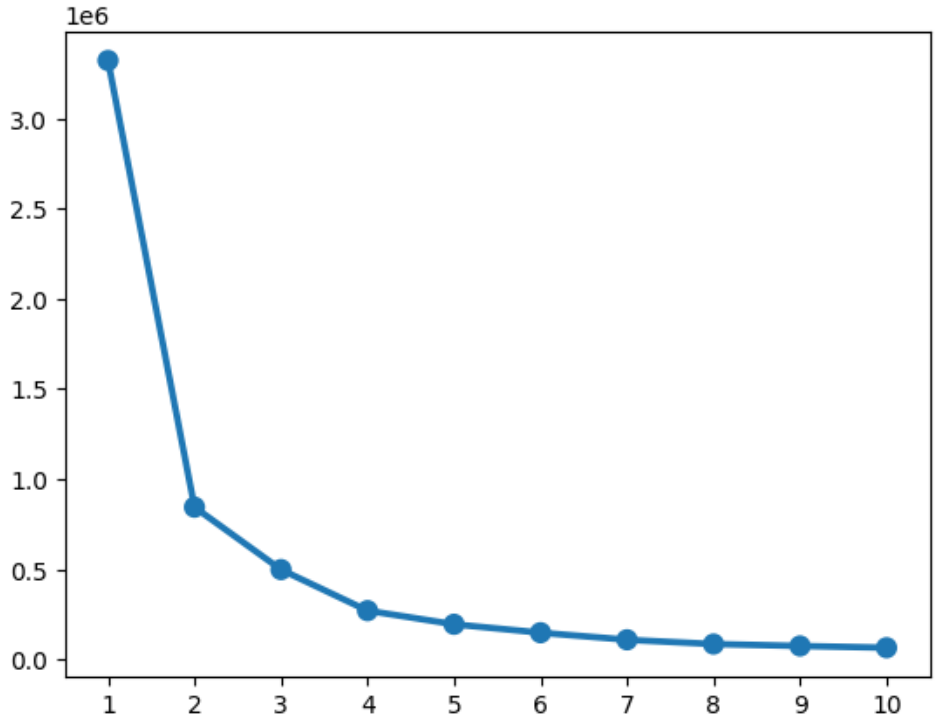
The elbow plot is a visualization technique to determine the optimal number of clusters in a clustering algorithm.

In an elbow plot, the number of clusters is plotted on the x-axis, and the corresponding "within-cluster sum of squares" (WCSS) or "inertia" is plotted on the y-axis. The WCSS is a measure of how internally coherent each cluster is.

The plot usually shows a downward curve where the WCSS decreases as the number of clusters increases. The "elbow" or "knee" of the curve represents a point where adding more clusters does not significantly reduce the WCSS. This elbow point is considered the optimal number of clusters for the given data set.

The rationale behind the elbow plot is to identify the point where the trade-off between the number of clusters and the quality of clustering is balanced. Too few clusters may result in underfitting, while too many clusters may lead to overfitting. The elbow plot helps in finding the "sweet spot" or the optimal number of clusters based on the specific problem and data set.

By analyzing the elbow plot, machine learning practitioners can make informed decisions about the appropriate number of clusters to use in their clustering algorithms, improving the overall effectiveness of the analysis.



Optimum number of clusters for k-means algorithm will be 3 since we can see between point 1 and 2, and 3 there is significant difference in the distance on Y axis but after that there is not significant difference between the distance points as they are very close to each other.

Cluster evaluation using SILHOUETTE SCORE

The silhouette score is a measure used to evaluate the quality of clustering results. It considers both the cohesion within clusters and the separation between clusters. The score ranges from -1 to 1, where higher scores indicate better-defined and well-separated clusters. A score close to 0 suggests overlapping or poorly separated clusters, while negative scores indicate potential misassignments.

The silhouette score is a useful tool for comparing different clustering algorithms or parameter settings and determining the optimal number of clusters.

```
The Average Silhouette Score for 2 clusters is 0.6485
The Average Silhouette Score for 3 clusters is 0.62221
The Average Silhouette Score for 4 clusters is 0.58781
The Average Silhouette Score for 5 clusters is 0.59934
The Average Silhouette Score for 6 clusters is 0.60913
The Average Silhouette Score for 7 clusters is 0.6033
The Average Silhouette Score for 8 clusters is 0.59729
The Average Silhouette Score for 9 clusters is 0.60811
The Average Silhouette Score for 10 clusters is 0.58838
The Average Silhouette Score for 11 clusters is 0.59464
The Average Silhouette Score for 12 clusters is 0.57357
The Average Silhouette Score for 13 clusters is 0.58474
The Average Silhouette Score for 14 clusters is 0.60099
```

- Silhouette score for a set of sample data points is used to measure how dense and well separated the clusters are.

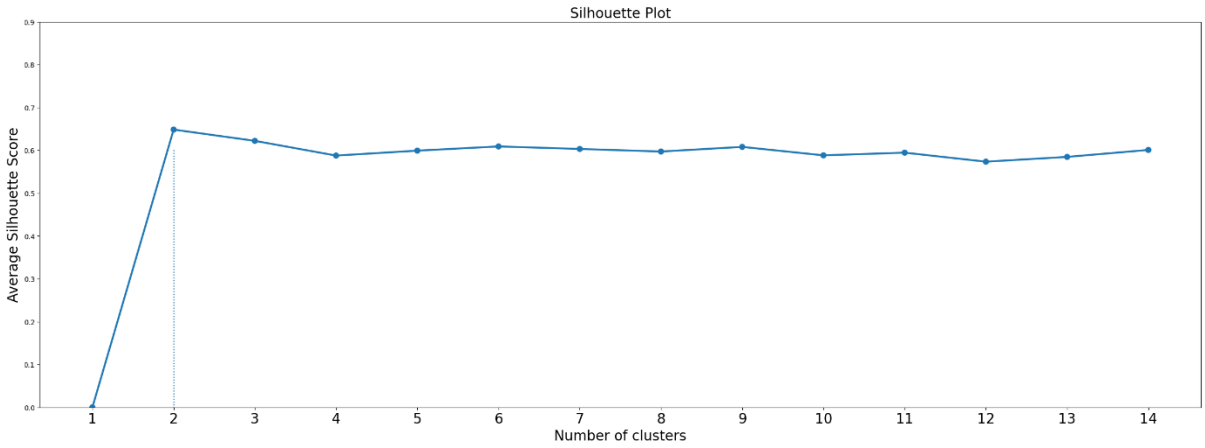
- Silhouette score takes into consideration the intra-cluster distance between the sample and other data points within the same cluster (a) and inter-cluster distance between the sample and the next nearest cluster (b).

- The silhouette score falls within the range [-1, 1].

- The silhouette score of 1 means that the clusters are very dense and nicely separated. The score of 0 means that clusters are overlapping. The score of less than 0 means that data belonging to clusters may be wrong/incorrect.

- The silhouette plots can be used to select the most optimal value of the K (no. of cluster) in K-means clustering.

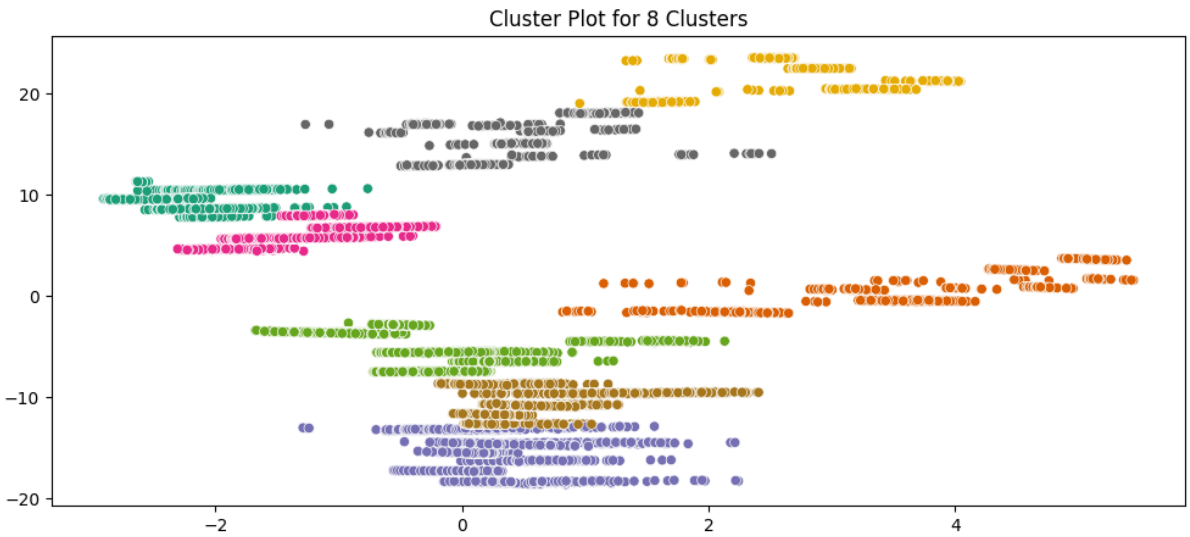
- The aspects to look out for in Silhouette plots are cluster scores below the average silhouette score, wide fluctuations in the size of the clusters, and also the thickness of the silhouette plot. The silhouette score is 0.6485



Clustering using K-MEANS

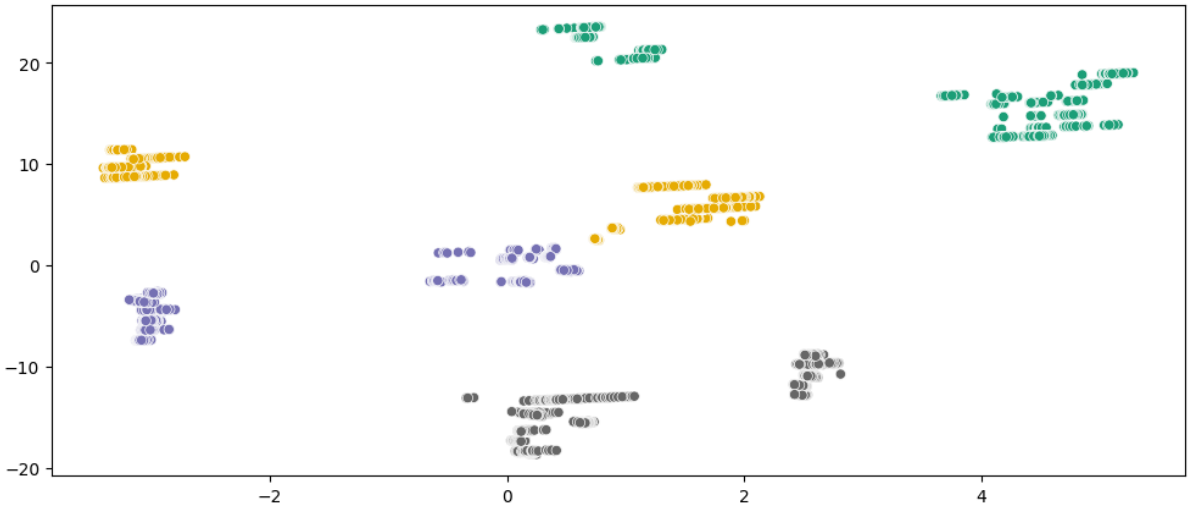
K-means clustering is an unsupervised machine learning algorithm used to group data points into k clusters. K is pre-defined. It iteratively assigns data points to the nearest centroid and updates the centroids until convergence. The algorithm aims to minimize the within-cluster sum of squares and create well-separated clusters. One challenge is determining the optimal number of clusters, k.

K-means clustering is widely used for tasks like customer segmentation and pattern recognition due to its efficiency, but it can be sensitive to initial centroid selection and outliers in the data.



PCA will reduce the dimensionality of the data to two dimensions.

For Ad_profiling, lets consider the number of Clusters 4



Clus_kmeans4	Clicks	Spend	Revenue	CTR	CPM	CPC	clusters	Clus_kmeans8	freq
0	0.205772	1.598414	1.529334	-0.990923	-0.989263	1.367987	5.445646	2.268769	3330
1	-0.485523	-0.618027	-0.587457	0.857577	0.839612	-0.693186	36.665102	1.944897	7241
2	0.759213	-0.189346	-0.203804	0.636611	0.581763	-0.774047	28.160459	5.709684	4967
3	-0.124942	0.012340	0.023032	-0.806587	-0.753852	0.572351	16.116100	4.939825	7528

CPM = (Total Campaign Spend / Number of Impressions) * 1,000

CPC = Total Cost (spend) / Number of Clicks

CTR = Total Measured Clicks / Total Measured Ad Impressions x 100

Cluster 0: This cluster exhibits the highest Click-Through Rate (CTR) and Cost Per Mille (CPM), indicating a high number of clicks relative to the number of impressions. Additionally, it has the lowest Cost Per Click (CPC), suggesting a lower cost for each click. The total cost is low, but the response in terms of clicks is high.

Cluster 1: In this cluster, the CTR and CPM are the lowest, indicating a low number of clicks compared to impressions. However, the Cost Per Click (CPC) is the highest, suggesting a higher cost for each click. The total cost is high, but the response in terms of clicks is lower.

Cluster 2: This cluster displays a high CTR and CPM, indicating a relatively high number of clicks compared to impressions. The Cost Per Click (CPC) is comparatively lower than in Cluster 1. The total cost is high, but the response in terms of clicks is lower.

Cluster 3: This cluster shows average values for CTR, CPM, and CPC. The cost and response (clicks) are at an average level.

Part 1: Clustering: Actionable Insights & Recommendations

- Extract meaningful insights (atleast 3) from the clusters to identify the most effective types of ads, target audiences, or marketing strategies that can be inferred from each segment.- Based on the clustering analysis and key insights, provide actionable recommendations (atleast 3) to Ads24x7 on how to optimize their digital marketing efforts, allocate budgets efficiently, and tailor ad content to specific audience segments.

RECOMMENDATIONS:

Spending:- Company needs to be focus on their spending part more, without spending in their ads they could not expect much clicks in their website.

Target audience by age group:- when click on ad gets increases then revenue is also increases. Company must be focus in which platform they showing their ads and which age -groups showing those particular side. Then company must be able to understood how they segmented their ads among youth or old people

Revenue:- for any company revenue is the most important part if company does not earned revenue they can not survive. Here, also from graphs and boxplot we can understand revenue is not that much high so they need to be working on their promotion part. Good qualities of ad and promotions is required .

Part 2: PCA: Define the problem and perform Exploratory Data Analysis

- Problem Definition- Check shape, Data types, statistical summary- Perform an EDA on the data to extract useful insights Note: 1. Pick 5 variables out of the given 24 variables below for EDA: No_HH, TOT_M, TOT_F, M_06, F_06, M_SC, F_SC, M_ST, F_ST, M_LIT, F_LIT, M_ILL, F_ILL, TOT_WORK_M, TOT_WORK_F, MAINWORK_M, MAINWORK_F, MAIN_CL_M, MAIN_CL_F, MAIN_AL_M, MAIN_AL_F, MAIN_HH_M, MAIN_HH_F, MAIN_OT_M, MAIN_OT_F 2. Example questions to answer from EDA- (i) Which state has highest gender ratio and which has the lowest? (ii) Which district has the highest & lowest gender ratio?

	State Code	Dist.Code	State	Area Name	No_HH	TOT_M	TOT_F	M_06	F_06	M_SC	...	MARG_CL_0_3_M	MARG_CL_0_3_F	MARG_AL_0_3_M	MARG_AL_0_
0	1	1	Jammu & Kashmir	Kupwara	7707	23388	29796	5862	6196	3	...	1150	749	180	
1	1	2	Jammu & Kashmir	Badgam	6218	19585	23102	4482	3733	7	...	525	715	123	
2	1	3	Jammu & Kashmir	Leh(Ladakh)	4452	6546	10964	1082	1018	3	...	114	188	44	
3	1	4	Jammu & Kashmir	Kargil	1320	2784	4206	563	677	0	...	194	247	61	
4	1	5	Jammu & Kashmir	Punch	11654	20591	29981	5157	4587	20	...	874	1928	465	

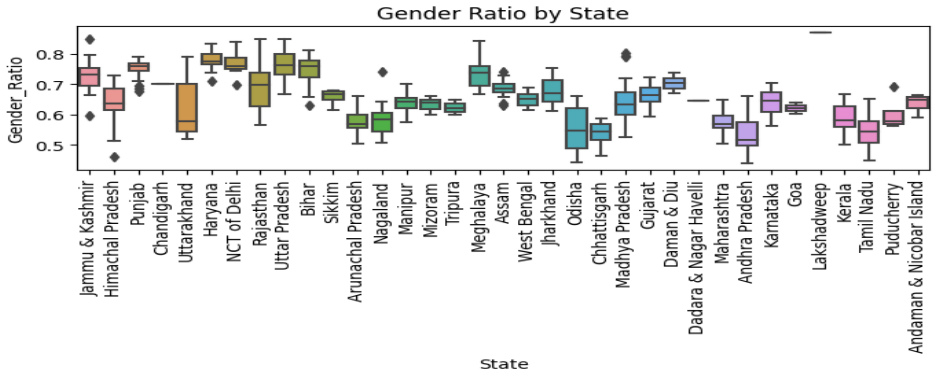
There are 640 rows and 61 columns in the data set.

There are no duplicated rows. No missing values are

there in this dataset.

(i) Which state has highest gender ratio and which has the lowest?

- state highest gender ratio: Lakshadweep
- state lowest gender ratio: Andhra Pradesh



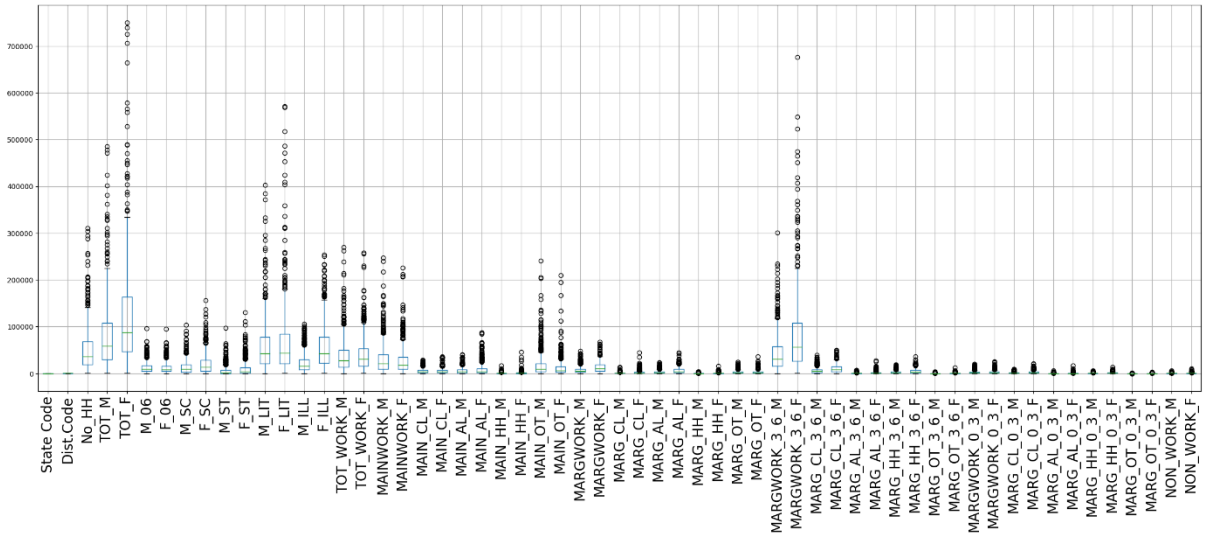
(ii) Which district has the highest & lowest gender ratio?

- area highest gender ratio: Lakshadweep
- area lowest gender ratio: Krishna

Part 2: PCA: Data Preprocessing

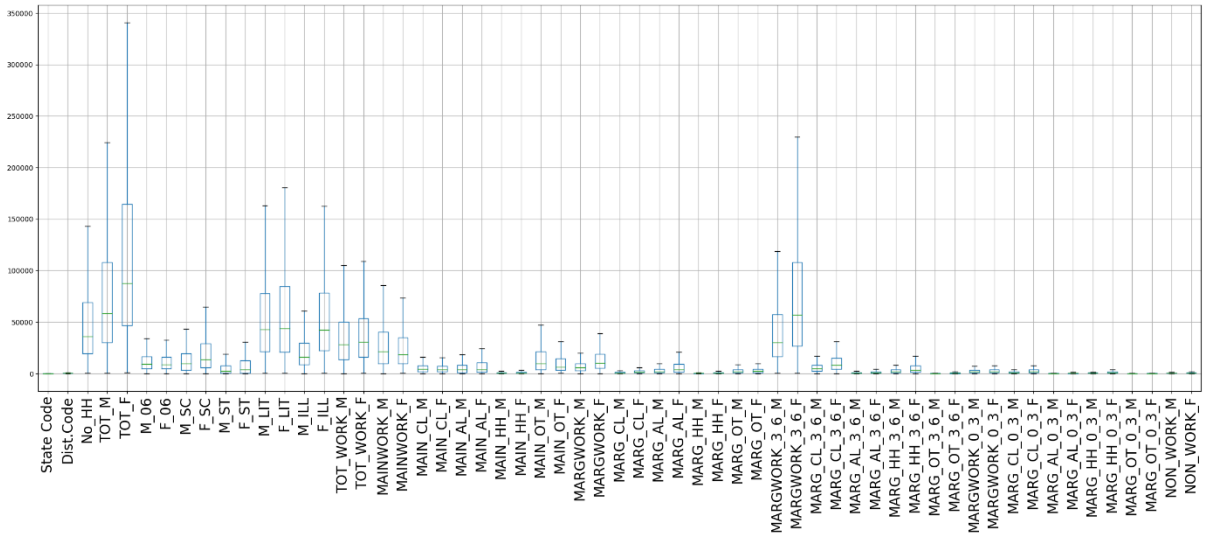
- Check for and treat (if needed) missing values- Check for and treat (if needed) data irregularities- Scale the Data using the z-score method- Visualize the data before and after scaling and comment on the impact on outliers.

BOXPLOT BEFORE TREATING THE OUTLIERS:



we will be removing outliers, since PCA is sensitive to outliers it might lead to inefficient results. Also the dataset has a lot of outliers which needs to be removed.

AFTER TREATING OUTLIERS THROUGH IQR:



So, firstly Scaling does not have an effect on outliers, scaling only reduces the magnitude of outliers but does not minimize them. we have applied zscore to the dataset to normalize it and to remove outliers, since we have seen that it has no impact on it, so we have used the iqr percentile function to remove outliers and now all the outliers are removed, now we can apply the zscore scaling technique again.

Part 2; PCA: PCA

- Create the covariance matrix- Get eigen values and eigen vectors- Identify the optimum number of PCs- Show Scree plot- Compare PCs with Actual Columns and identify which is explaining most variance- Write inferences about all the PCs in terms of actual variables- Write linear equation for first PC Note: For the scope of this project, take at least 90% explained variance.

COVARIANCE MATRIX:

```
9] pd.DataFrame(np.round(pca.get_covariance(),2),columns=pca_scaled.columns,index=pca_scaled.columns) #cov matrix
```

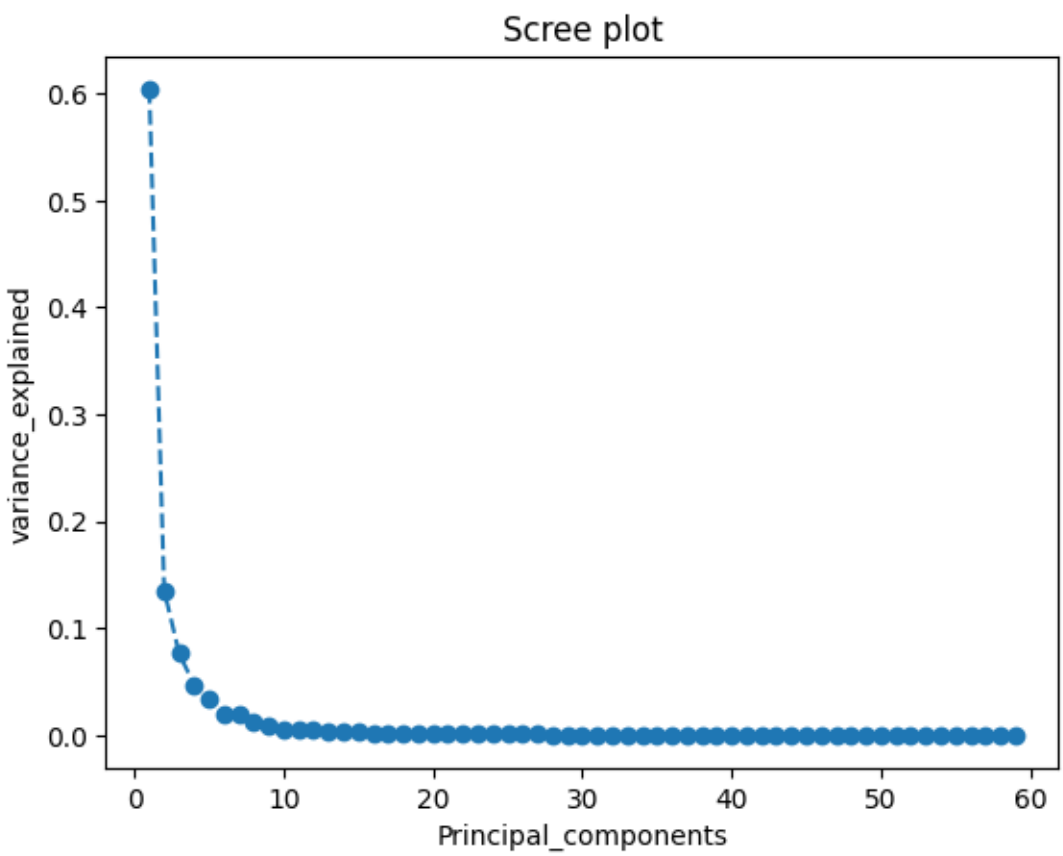
MARG_CL_M	-0.28	-0.28	0.23	0.37	0.32	0.48	0.48	0.30	0.27	0.11	0.09	0.32	0.24	0.46	0.41	0.26	0.26	0.13
MARG_CL_F	-0.26	-0.27	0.14	0.23	0.21	0.33	0.33	0.18	0.16	0.15	0.13	0.20	0.15	0.29	0.29	0.14	0.22	0.04
MARG_AL_M	0.00	0.03	0.51	0.60	0.57	0.67	0.69	0.56	0.55	0.15	0.16	0.52	0.42	0.73	0.72	0.52	0.49	0.39
MARG_AL_F	0.26	0.28	0.58	0.52	0.57	0.54	0.55	0.48	0.53	0.35	0.37	0.47	0.41	0.58	0.72	0.48	0.65	0.38
MARG_HH_M	-0.07	-0.05	0.59	0.76	0.70	0.80	0.81	0.70	0.66	-0.03	-0.05	0.71	0.61	0.81	0.75	0.68	0.52	0.58
MARG_HH_F	0.08	0.08	0.66	0.73	0.72	0.72	0.73	0.67	0.67	0.02	0.01	0.71	0.64	0.74	0.74	0.71	0.62	0.62
MARG_OT_M	0.14	0.13	0.80	0.90	0.87	0.86	0.85	0.78	0.76	0.03	0.01	0.90	0.87	0.81	0.77	0.89	0.70	0.83
MARG_OT_F	0.24	0.22	0.84	0.86	0.87	0.80	0.79	0.74	0.75	0.11	0.10	0.88	0.87	0.76	0.77	0.87	0.76	0.84
MARGWORK_3_6_M	0.08	0.08	0.86	0.98	0.95	0.99	0.98	0.86	0.83	0.01	0.00	0.96	0.89	0.95	0.91	0.92	0.77	0.87
MARGWORK_3_6_F	0.16	0.16	0.91	0.99	0.98	0.95	0.94	0.87	0.85	0.00	-0.00	0.99	0.95	0.90	0.89	0.96	0.79	0.93
MARG_CL_3_6_M	0.06	0.06	0.73	0.85	0.82	0.88	0.89	0.74	0.72	0.10	0.09	0.81	0.72	0.89	0.85	0.79	0.67	0.67
MARG_CL_3_6_F	0.24	0.24	0.78	0.76	0.80	0.76	0.77	0.67	0.70	0.29	0.30	0.73	0.68	0.76	0.86	0.73	0.82	0.64
MARG_AL_3_6_M	-0.24	-0.24	0.26	0.41	0.36	0.51	0.52	0.33	0.30	0.13	0.10	0.35	0.27	0.50	0.45	0.30	0.29	0.17

EIGEN VALUES AND EIGEN VECTORS:

```
array([3.57108475e+01, 7.98557733e+00, 4.50785903e+00, 2.77867519e+00,
       1.97472860e+00, 1.17776767e+00, 1.13039501e+00, 7.22103375e-01,
       4.64431676e-01, 3.46774532e-01, 3.05963732e-01, 2.68366978e-01,
       2.20811847e-01, 1.80278141e-01, 1.68296796e-01, 1.32409265e-01,
       1.29436740e-01, 1.03406138e-01, 9.55347371e-02, 8.58417456e-02,
       8.09066019e-02, 6.56476263e-02, 6.23708292e-02, 4.79008765e-02,
       4.56408231e-02, 4.38435424e-02, 3.10290046e-02, 2.86009130e-02,
       2.74987147e-02, 2.33916183e-02, 2.16432655e-02, 1.87723745e-02,
       1.56678899e-02, 1.40371782e-02, 1.18761437e-02, 1.11316049e-02,
       9.08077540e-03, 7.25913797e-03, 6.18691864e-03, 4.89879738e-03,
       4.55034891e-03, 4.24001897e-03, 3.26372660e-03, 2.18239672e-03,
       2.12902353e-03, 1.90742071e-03, 1.43490578e-03, 1.09833856e-03,
       9.62038195e-04, 8.56614567e-04, 6.51562449e-04, 5.76295579e-04,
       4.60548161e-05, 8.95049889e-05, 1.38262667e-04, 2.07171377e-04,
       4.31846786e-04, 3.69015469e-04, 3.06582238e-04])
```

```
array([[-0.03036889, -0.17219661, -0.30134696, ...,  0.02156452,
         0.00500425,  0.00171484],
       [-0.03034403, -0.1690169 , -0.30641795, ..., -0.02250001,
        -0.00665917, -0.00213945],
       [-0.14957826, -0.11969182, -0.06824342, ..., -0.02710355,
         0.01252643,  0.02384027],
       ...,
       [-0.14106815,  0.04130293,  0.07350673, ...,  0.03029966,
         0.02554542,  0.02740047],
       [-0.14743391, -0.03753818,  0.11220488, ..., -0.01073912,
        -0.00174554,  0.01358457],
       [-0.14214317, -0.03868258,  0.01794257, ...,  0.00069812,
        -0.01051117, -0.0014845 ]])
```

SCREE PLOT:



Although there are 60 observed variables the first 5 principal components can explain more than 90% of total variation, hence the optimum number of component's that we take is 5 instead of original 60, thereby reducing the dimensions to a significant number. We also lookat the scree plot which has indices of the

number of component on the X-axis and variances on the Y-axis. We can see that at point 5 there is a distinct break point and the line joining the variances (elbow pint) becomes approximately horizontal.

Compare PCs with Actual Columns and identify which is explaining most variance-Write inferences about all the PCs in terms of actual variables the names we give to PC's are:

PC1= Skilled_unskilled_workforce

PC2= Marginalized_working_population

PC3= Female_workforce

PC4= Household_population

PC5= SC/ST workforce

Write linear equation for first PC Note: For the scope of this project, take at least 90%explained variance.

```
Linear equation for the first PC: PC1 = 0.03 * State Code + 0.03 * Dist
.Code + 0.15 * No_HH + 0.16 * TOT_M + 0.16 * TOT_F + 0.16 * M_06 + 0.16
* F_06 + 0.14 * M_SC + 0.14 * F_SC + 0.02 * M_ST + 0.02 * F_ST + 0.16 *
M_LIT + 0.15 * F_LIT + 0.15 * M_ILL + 0.16 * F_ILL + 0.15 * TOT_WORK_M
+ 0.14 * TOT_WORK_F + 0.14 * MAINWORK_M + 0.13 * MAINWORK_F + 0.11 * MA
IN_CL_M + 0.08 * MAIN_CL_F + 0.12 * MAIN_AL_M + 0.09 * MAIN_AL_F + 0.14
* MAIN_HH_M + 0.13 * MAIN_HH_F + 0.12 * MAIN_OT_M + 0.12 * MAIN_OT_F +
0.16 * MARGWORK_M + 0.15 * MARGWORK_F + 0.09 * MARG_CL_M + 0.06 * MARG_
CL_F + 0.13 * MARG_AL_M + 0.12 * MARG_AL_F + 0.14 * MARG_HH_M + 0.14 *
MARG_HH_F + 0.15 * MARG_OT_M + 0.15 * MARG_OT_F + 0.16 * MARGWORK_3_6_M
+ 0.16 * MARGWORK_3_6_F + 0.16 * MARG_CL_3_6_M + 0.15 * MARG_CL_3_6_F +
0.09 * MARG_AL_3_6_M + 0.07 * MARG_AL_3_6_F + 0.13 * MARG_HH_3_6_M + 0.
11 * MARG_HH_3_6_F + 0.14 * MARG_OT_3_6_M + 0.14 * MARG_OT_3_6_F + 0.15
* MARGWORK_0_3_M + 0.15 * MARGWORK_0_3_F + 0.14 * MARG_CL_0_3_M + 0.13
* MARG_CL_0_3_F + 0.06 * MARG_AL_0_3_M + 0.06 * MARG_AL_0_3_F + 0.12 *
MARG_HH_0_3_M + 0.11 * MARG_HH_0_3_F + 0.14 * MARG_OT_0_3_M + 0.14 * MA
RG_OT_0_3_F + 0.15 * NON_WORK_M + 0.14 * NON_WORK_F
```

THANKYOU